IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

MUKUL
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
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Executive Summary

- **Summary of methodologies:-**
- This project followed a structured data science approach to predict Falcon 9 first-stage landing success.
- First, data was collected using the SpaceX REST API and supplemented with web scraping techniques. The raw data was then cleaned and preprocessed through data wrangling, handling missing values, and selecting relevant features.
- Next, SQL queries and exploratory data analysis (EDA) were performed to identify trends and relationships between variables such as payload mass, launch site, booster version, and landing outcome. Visualizations were created to better understand these patterns.
- Finally, multiple machine learning models were implemented, including Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). The models were trained and evaluated to determine the most accurate predictor of landing success.

- **Summary of all results:-**
- The analysis revealed that **payload mass, booster version category, and launch site** are key factors influencing Falcon 9 first-stage landing success.
- Launch sites such as KSC LC-39A and CCAFS LC-40 demonstrated comparatively higher success rates, indicating operational efficiency differences across locations.
- A clear relationship was observed between **payload mass and landing outcome**, showing that mission characteristics directly affect recovery performance.
- Machine Learning models successfully learned patterns from historical launch data, with the best-performing model achieving strong predictive accuracy.
- The interactive dashboard further confirmed these findings by allowing dynamic exploration of success rates across payload ranges and launch sites.
- Overall, the results demonstrate that **data-driven modeling can effectively predict rocket landing success**, supporting better mission planning and cost optimization strategies.

Introduction

- SpaceX has revolutionized the aerospace industry by introducing reusable rocket technology, significantly reducing the cost of space missions. The Falcon 9 rocket is designed to land and recover its first stage after launch, allowing it to be reused in future missions. This innovation plays a crucial role in improving operational efficiency and lowering overall launch expenses.
- However, not every launch results in a successful landing. The success of the first-stage landing depends on several factors such as payload mass, launch site, booster version, and mission profile. Accurately predicting landing outcomes is important for cost estimation, risk assessment, and mission planning.
- This project aims to analyze historical Falcon 9 launch data and develop a machine learning model to predict whether the first stage will successfully land. By applying data analytics and predictive modeling techniques, we seek to provide data-driven insights that support better decision-making in aerospace operations.

Section 1

Methodology

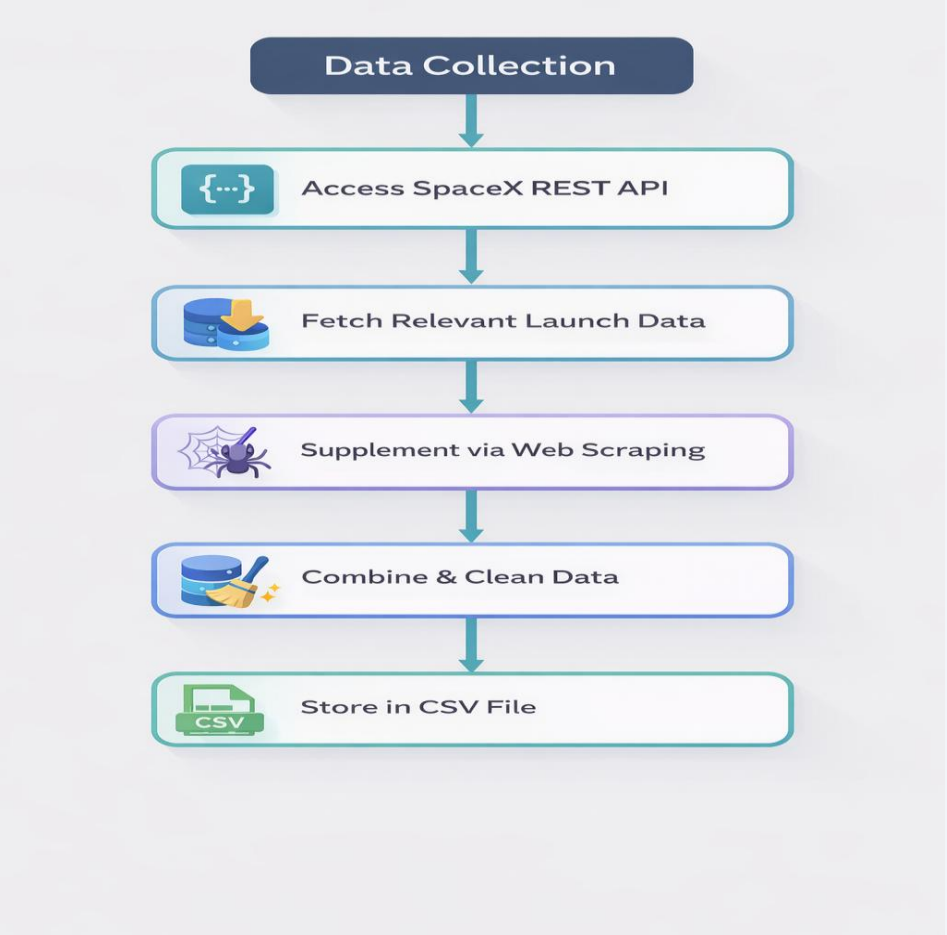
Methodology

Executive Summary

- Data collection methodology:
  - Launch data was collected using the **SpaceX REST API**, which provides structured historical launch records using ***GET Method.***
  - Additional data was gathered using **web scraping techniques** to supplement missing or specific launch-related information.
- Perform data wrangling
  - Cleaned and transformed the collected launch data by handling missing values, selecting relevant features, converting variables into appropriate formats, and preparing the dataset for analysis and machine learning modeling.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Built classification models on preprocessed data, tuned them using cross-validation and hyperparameter optimization, and evaluated their performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrix to select the best predictive model.

Data Collection

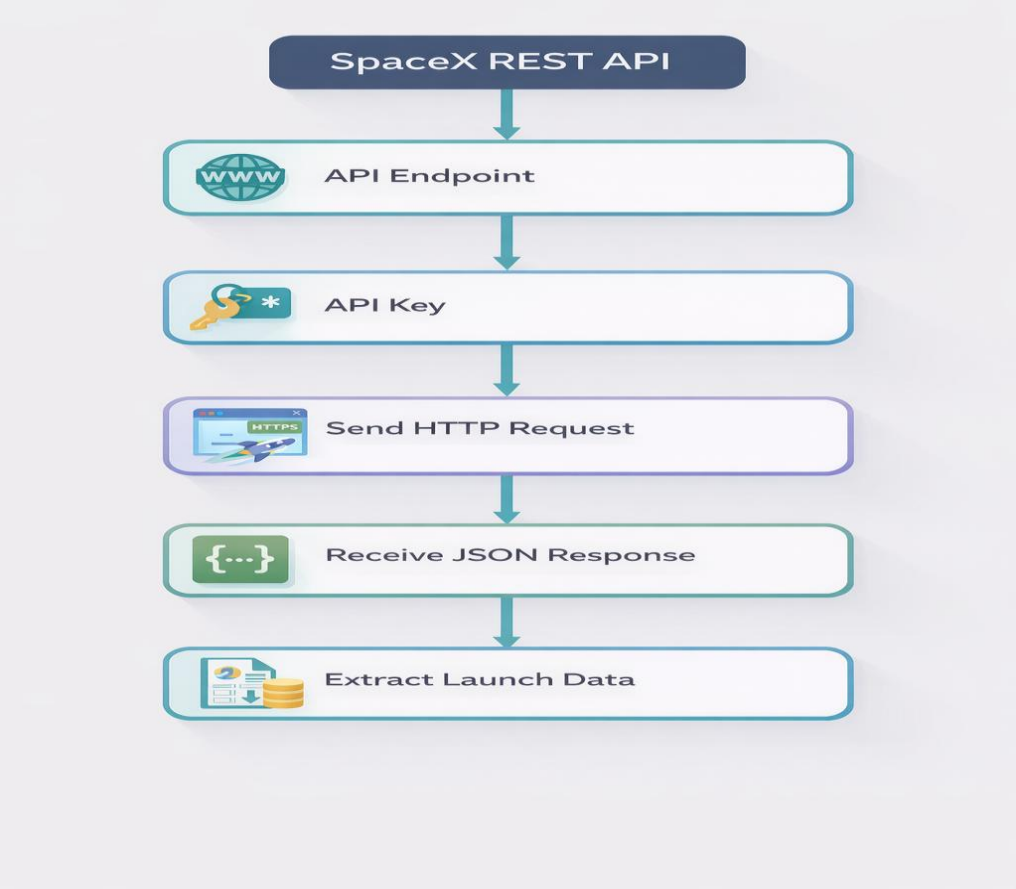
•Launch data was collected using the **SpaceX REST API**, which provides structured historical launch records.
•Relevant information such as flight number, launch site, payload mass, booster version, orbit type, and landing outcome was extracted
•. Additional data was gathered using **web scraping techniques** to supplement missing or specific launch-related information.
•The collected data was stored in CSV format for further processing and analysis.



Data Collection – SpaceX API

•Launch data was collected using the **SpaceX REST API**, which provides structured historical launch records.
•Relevant information such as flight number, launch site, payload mass, booster version, orbit type, and landing outcome was extracted.

- View on GitHub: https://github.com/mukul816/Spacex-project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



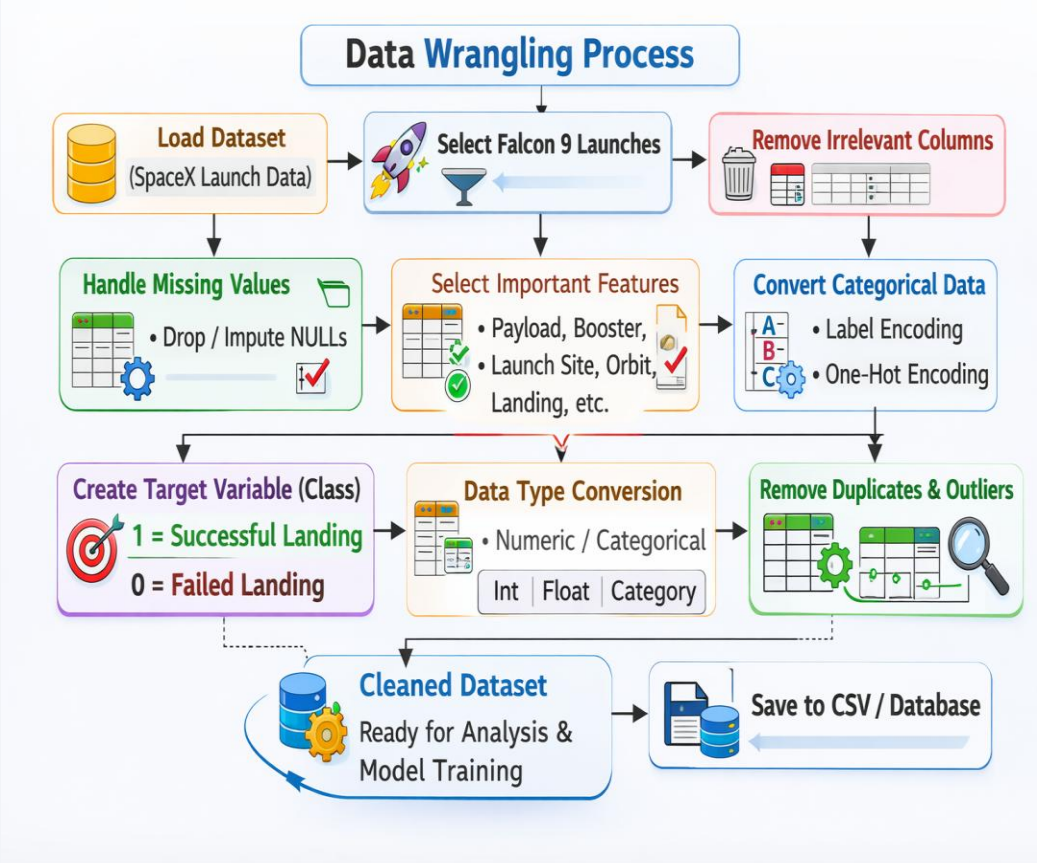
Data Collection - Scraping

•Data was gathered using **web scraping techniques** to supplement missing or specific launch-related information.
•The collected data was stored in CSV format for further processing and analysis

•**View on GitHub:-**
•http://github.com/mukul816/Spacex-project/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

- **Here is how data was wrangled:-**
- Filtered the dataset to include only **Falcon 9 launches**.
- Removed irrelevant columns and retained important features such as payload mass, booster version, launch site, orbit, and landing outcome.
- Handled missing values by dropping incomplete records or replacing null values where necessary.
- Converted categorical variables into numerical format for machine learning modeling.
- Created a binary target variable (**Class**) where:
  - 1 = Successful Landing
  - 0 = Failed Landing
- Ensured correct data types and formatted the dataset for analysis and model training.



- **View on GitHub:**
- https://github.com/mukul816/Spacex-project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- **Summary of Charts Used in EDA:-**

- **1. Flight Number vs Landing Outcome (Scatter Plot)**
  - **What it shows:** Landing success across increasing flight numbers.
  - **Why used:** To observe improvement trends over time and assess whether experience increases success probability.
- . **2. Payload Mass vs Landing Outcome (Scatter Plot)**
  - **What it shows:** Relationship between payload mass and landing success/failure.
  - **Why used:** To examine whether heavier payloads affect landing performance and detect patterns influencing prediction.
- **3. Orbit Type vs Success (Bar Chart)**
  - **What it shows:** Success rate across different orbit categories.
  - **Why used:** To analyze how mission type impacts landing outcome.
- **4. Launch Success Count (Bar Chart)**
  - **What it shows:** Number of successful vs failed landings.
  - **Why used:** To understand overall class distribution and check whether the dataset is balanced.
- **5. Launch Site vs Success Rate (Bar Chart)**
  - **What it shows:** Success count grouped by launch site.
  - **Why used:** To identify which launch sites have higher landing success rates and detect operational differences.
- **6. Booster Version Category vs Success (Bar Chart)**
  - **What it shows:** Success distribution across booster types.
  - **Why used:** To determine whether certain booster versions perform better.

- **View on GitHub:**
- https://github.com/mukul816/Spacex-project/blob/main/edadataviz.ipynb

EDA with SQL

- Queried the database to retrieve key launch information such as flight number, launch site, payload mass, orbit type, and landing outcome.
- Calculated the total number of launches and identified successful versus failed landings using aggregation functions.
- Grouped data by launch site to determine which locations had higher landing success rates.
- Analyzed mission distribution across different orbit types to understand their impact on landing performance.
- Used SQL filtering and sorting techniques to identify trends, patterns, and performance variations.
- Extracted structured insights from raw launch data to support visualization and machine learning modeling.

- **View on GitHub:**
- **https://github.com/mukul816/Spacex-project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb**

Build an Interactive Map with Folium

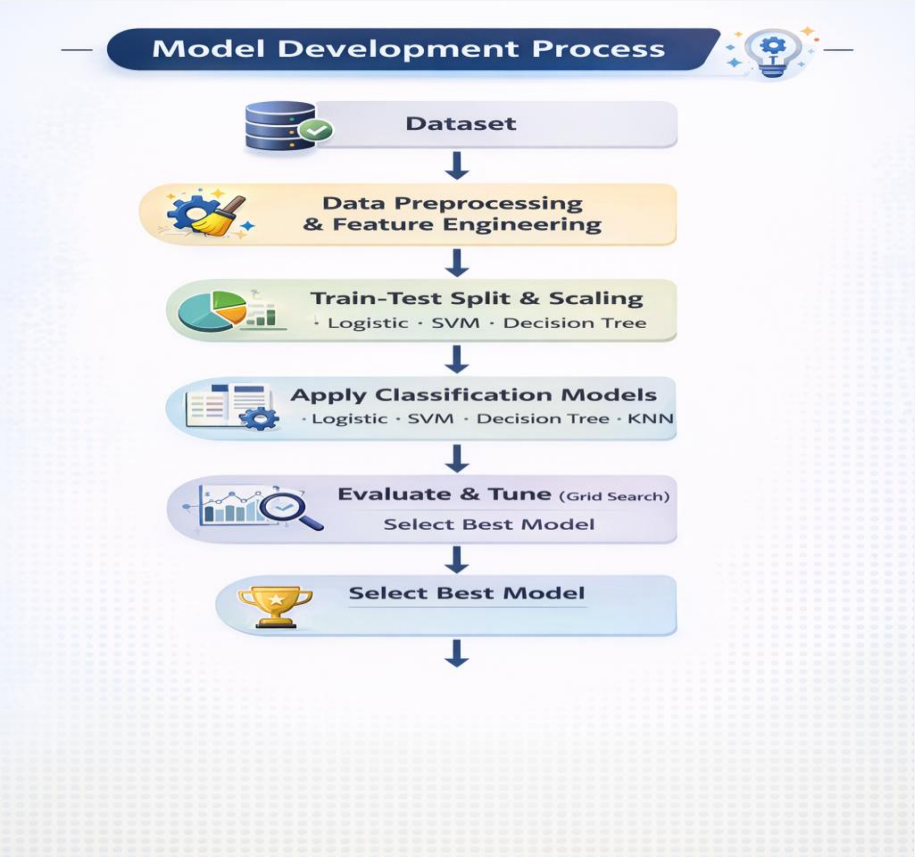
- Added **markers** to represent each SpaceX launch site location on the map.
  → This helped visually identify the geographical distribution of launch sites.
- Used **circles (CircleMarker)** to highlight launch sites with different sizes or colors based on launch outcomes.
  → This made it easier to compare success rates across locations.
- Plotted **lines (PolyLine)** between launch sites and nearby landmarks (such as coastlines, railways, or highways).
  → This helped measure distances and understand geographical advantages influencing landing operations.
- Included **pop-up information boxes** displaying launch site details.
  → This provided interactive insights when users clicked on map elements.
- Applied different **colors for success and failure outcomes**.
  → This improved visual interpretation of landing performance across sites.
- **These map objects were used to:-**
- Visually analyze geographical patterns
- Understand the spatial relationship between launch sites and surroundings
- Identify potential environmental or location-based factors affecting landing success
- Provide interactive and intuitive geographic insights
- **View on GitHub:**
- https://github.com/mukul816/Spacex-project/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- **Dashboard Visualizations & Interactions:-**
- **Plots Added:-**
- **Pie Chart (Success Distribution by Launch Site)**
  → Displays total successful launches for all sites or success vs failure for a selected site.
  → Helps quickly compare landing performance across different launch locations.
- **Scatter Plot (Payload Mass vs Landing Outcome)**
  → Shows the relationship between payload mass and landing success.
  → Colored by booster version category to identify performance differences.
  → Helps detect patterns and correlations affecting landing outcomes.
- **Interactive Components Added:-**
- **Launch Site Dropdown Filter**
  → Allows users to select “All Sites” or a specific launch site.
  → Enables dynamic comparison of performance across different locations.
- **Payload Range Slider**
  → Filters data based on selected payload mass range.
  → Helps analyze how different payload weights influence landing success.
- **These Plots & Interactions Were Added:-**
- To allow **interactive exploration** of launch performance.
- To identify **key factors influencing landing success**.
- To make insights more intuitive and visually understandable.
- To support decision-making through dynamic filtering and real-time visualization.

Predictive Analysis (Classification)

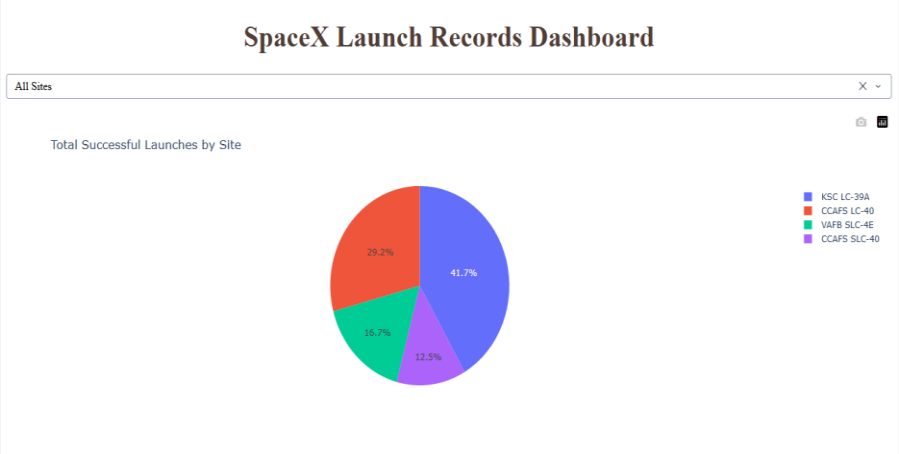
- **Model Development Process:-**
- Prepared and cleaned the dataset, selected features, and performed train-test split.
- Applied feature scaling to standardize numerical variables.
- Built classification models: Logistic Regression, SVM, Decision Tree, and KNN.
- Evaluated models using accuracy, cross-validation, and confusion matrix.
- Tuned hyperparameters using Grid Search to improve performance.
- Compared results and selected the best-performing model.



- **View on GitHub:-**
- **https://github.com/mukul816/Spacex-project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb**

Results

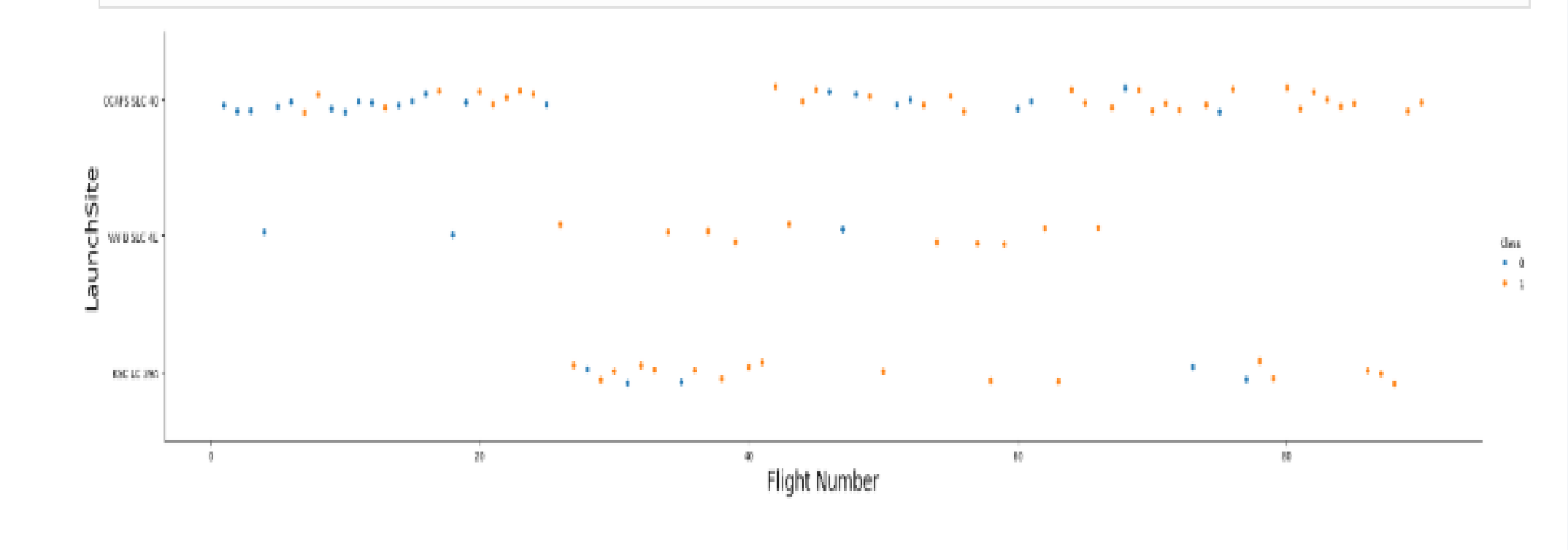
- **EDA Results:-**
  - Landing success improved over time with increasing flight numbers.
  - Launch site, payload mass, booster version, and orbit type significantly influenced landing outcomes.
  - Certain launch sites and booster versions showed higher success rates.
  - Key variables identified during EDA were used for predictive modeling.
- Predictive analysis results:-
  - Multiple classification models were developed to predict landing success.
  - After evaluation and hyperparameter tuning, the best-performing model achieved strong predictive accuracy.
  - Feature scaling and cross-validation improved model reliability.
  - Payload mass, launch site, and booster version were the most influential predictors.
  - The final model successfully demonstrated that landing outcomes can be predicted using historical launch data.



Section 2

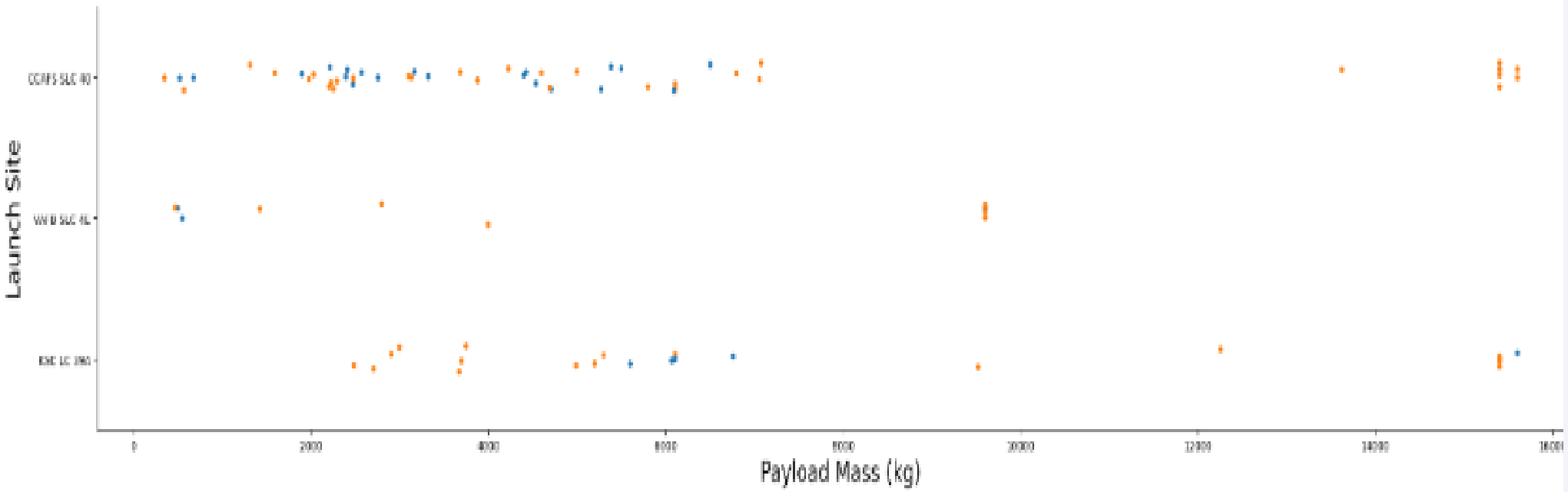
Insights drawn from EDA

Flight Number vs. Launch Site



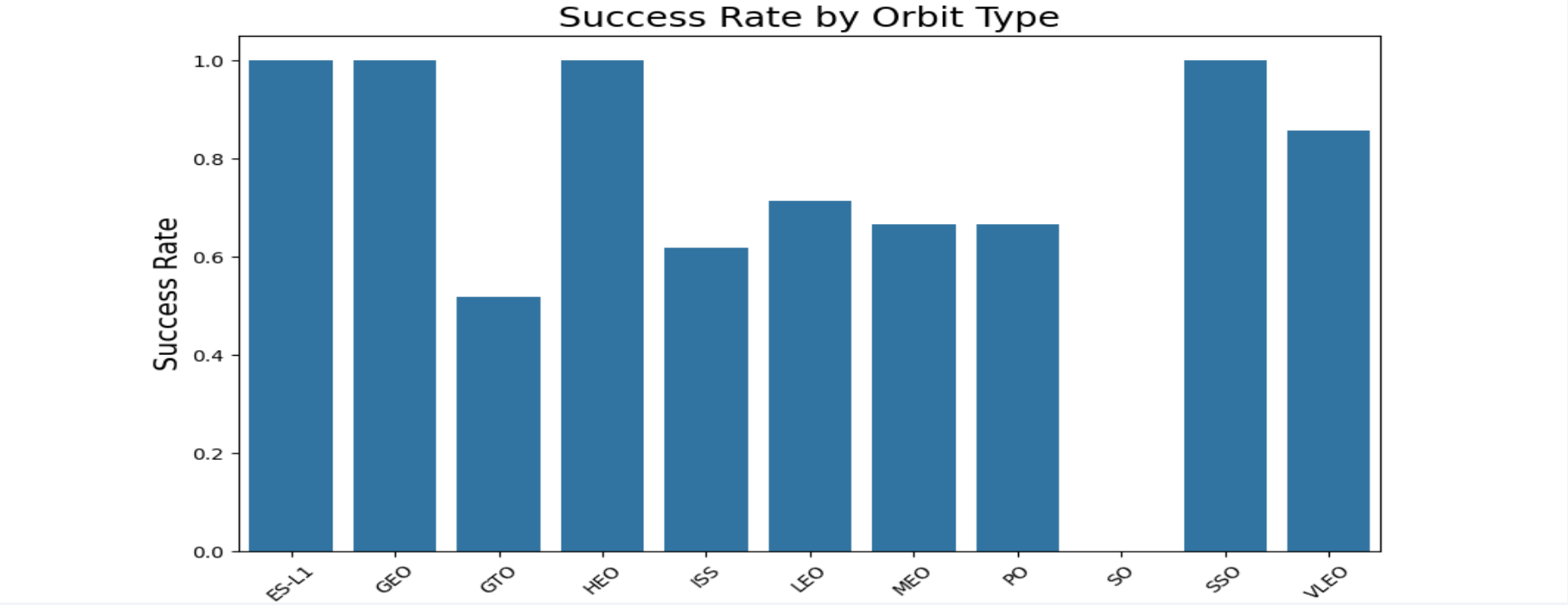
- Success rate improved with increasing flight numbers, indicating learning and technological advancement and CCAFS SLC 40 has more success rate.

Payload vs. Launch Site



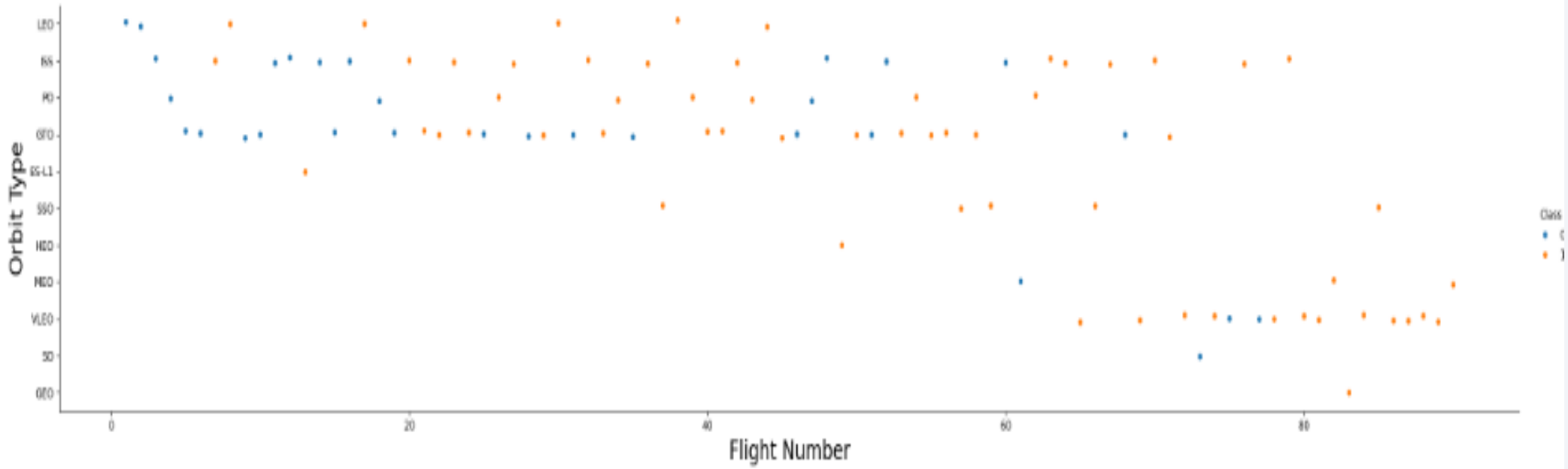
Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type



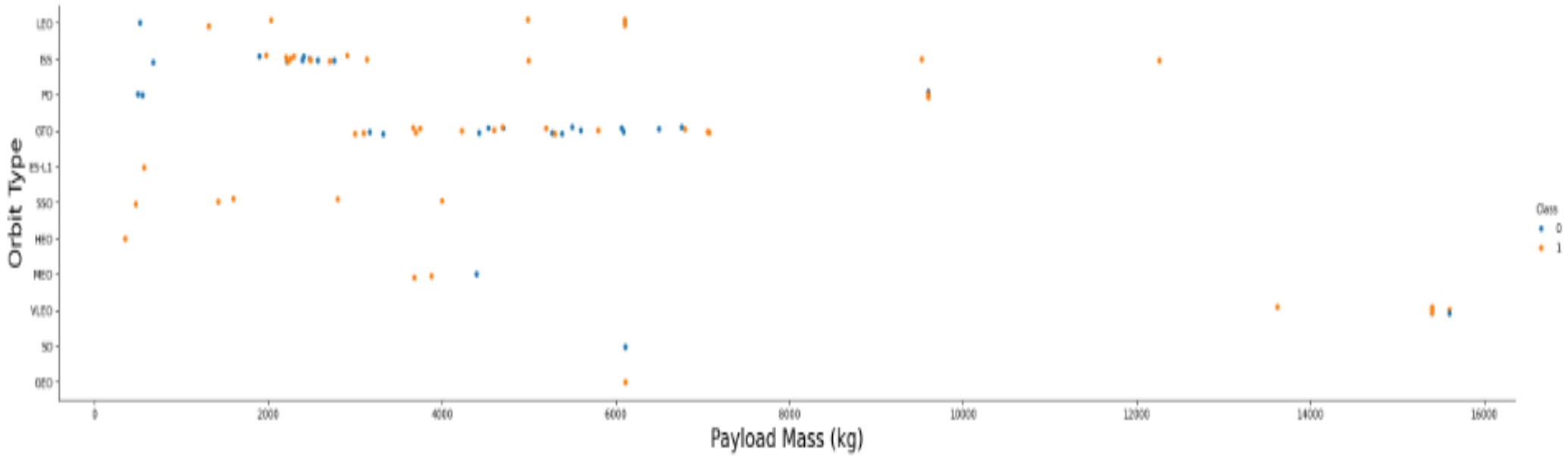
- ES-L1, GEO, HEO, and SSO orbits all have a high success rate.

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

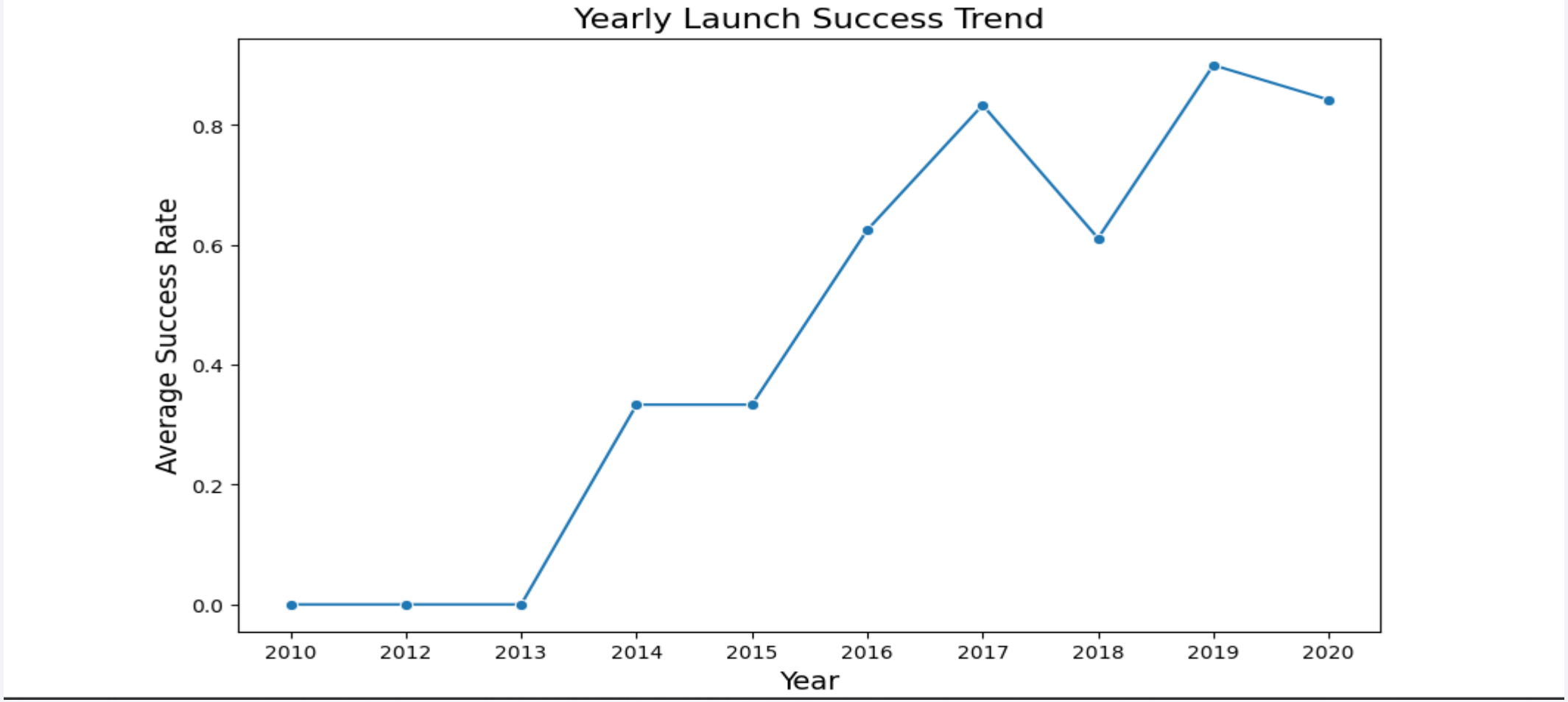
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



- you can observe that the success rate since 2013 kept increasing till 2020.

All Launch Site Names

- Here is all-unique launch sites with SQL query:-

```
In [10]: %sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

 * sqlite:///my_data1.db
Done.
```

Out[10]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Here is the 5 records where launch sites begin with the string 'CCA':-

In [13]:

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcor
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachu
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachu
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No atten
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No atten
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No atten

Total Payload Mass

In [14]:
```
%%sql SELECT SUM(Payload_Mass__kg_) AS Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

Total_Payload_Mass
45596

- The total payload mass carried by boosters launched by NASA (CRS) is equal to 45596.

Average Payload Mass by F9 v1.1

In [15]:
```
%%sql SELECT AVG(Payload_Mass__kg_) AS Avg_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[15]:

Avg_Payload_Mass
2928.4

- Average payload mass carried by booster version F9 v1.1 is equal to 2928.4.

First Successful Ground Landing Date

In [16]:

```
%%sql SELECT MIN(Date) AS First_Success_Ground_Pad
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

First_Success_Ground_Pad
2015-12-22

- The first successful landing outcome in ground pad was achieved on 2015-12-22.

Successful Drone Ship Landing with Payload between 4000 and 6000

- The boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000:-

In [17]:

```
%%sql SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND Payload_Mass__kg_ > 4000
AND Payload_Mass__kg_ < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes:-

In [18]:

```
%%sql SELECT Mission_Outcome, COUNT(*) AS Total_Count
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- All the booster versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function are:-

In [19]:

```
%%sql SELECT Booster_Version
FROM SPACEXTABLE
WHERE Payload_Mass__kg_ = (
    SELECT MAX(Payload_Mass__kg_)
    FROM SPACEXTABLE
);
```

Out[19]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List of the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015 are:-

In [20]:

```
%%sql SELECT
    substr(Date, 6, 2) AS Month,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE substr(Date, 0, 5) = '2015'
AND Landing_Outcome = 'Failure (drone ship)';
```

Out[20]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank of the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order are:-

In [21]:

```
%%sql SELECT
    Landing_Outcome,
    COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

Out[21]:

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

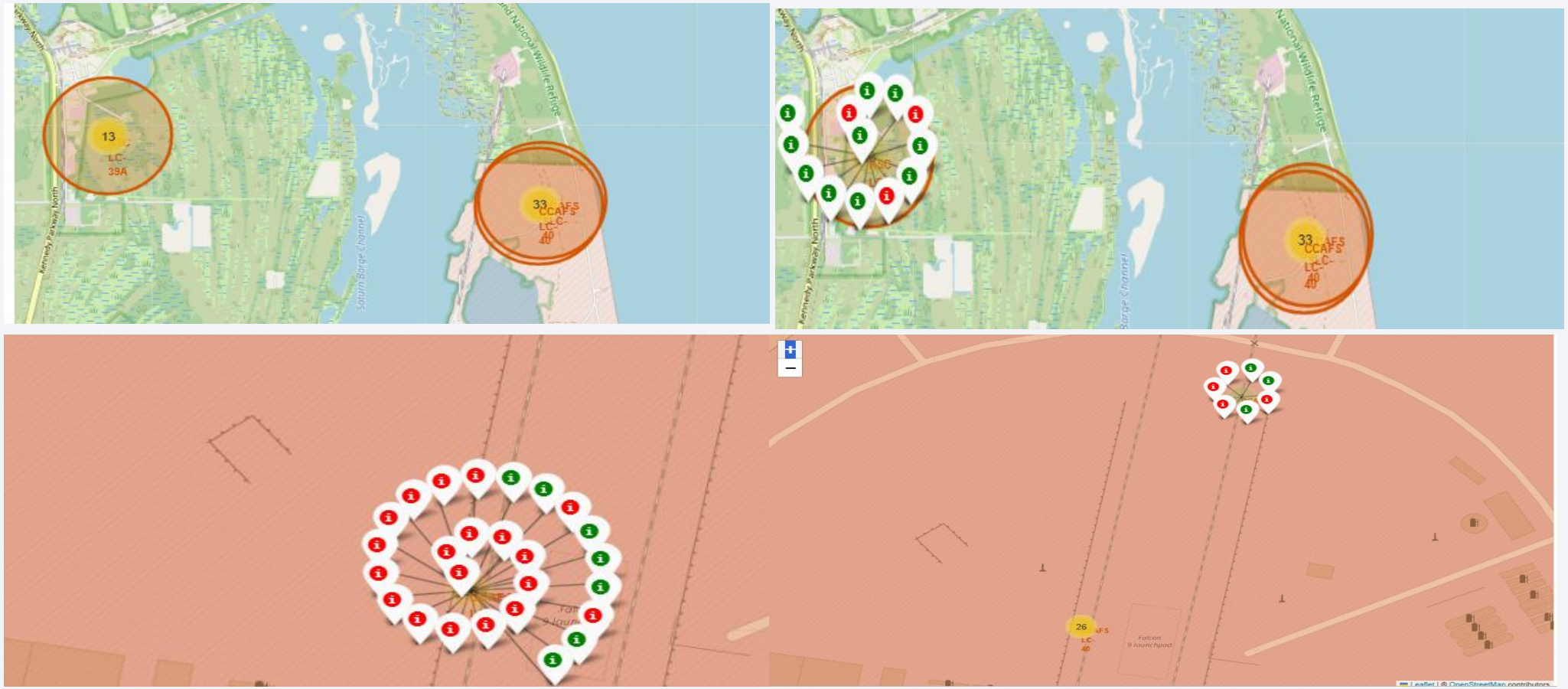
Section 3

Launch Sites Proximities Analysis

All launch sites on a map

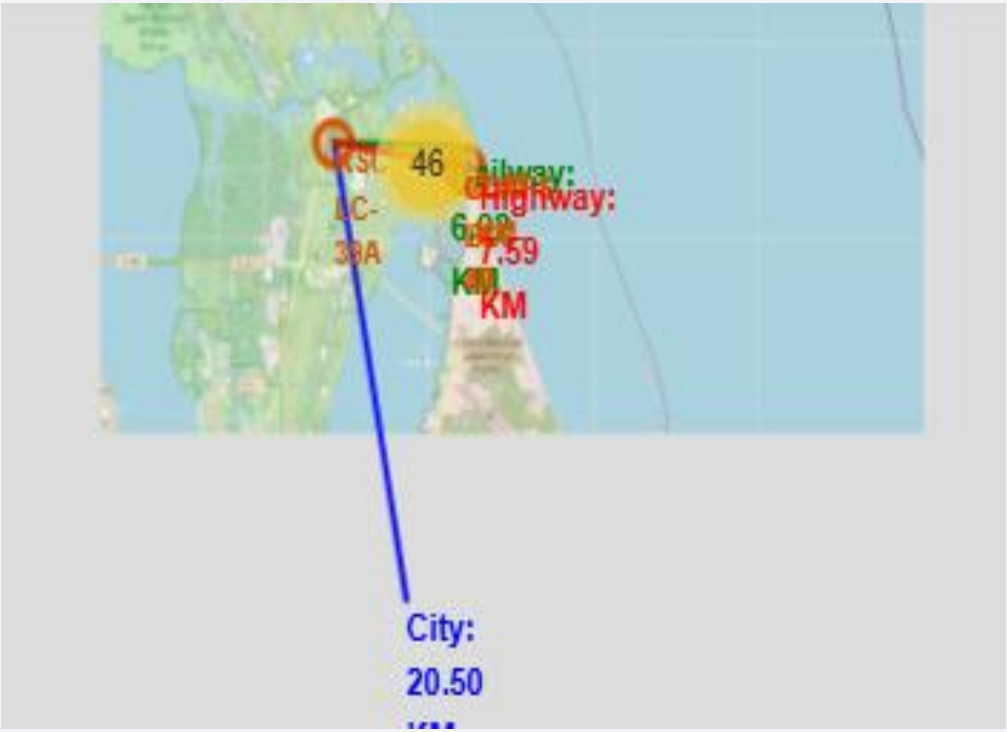
- No, not all launch sites are in proximity to the **Equator.**
- Most launch sites are in close proximity to the **coast** so that if a failure occurs during ascent, debris will fall into the **ocean** instead of populated areas.

Success/failed launches for each site on the map



- The Marker Cluster visualization shows that **KSC LC-39A has the highest success rate**, while **CCAFS SLC-40 has the highest volume of launches** with a more even split between successes (green) and failures (red).

The distances between a launch site to its proximities

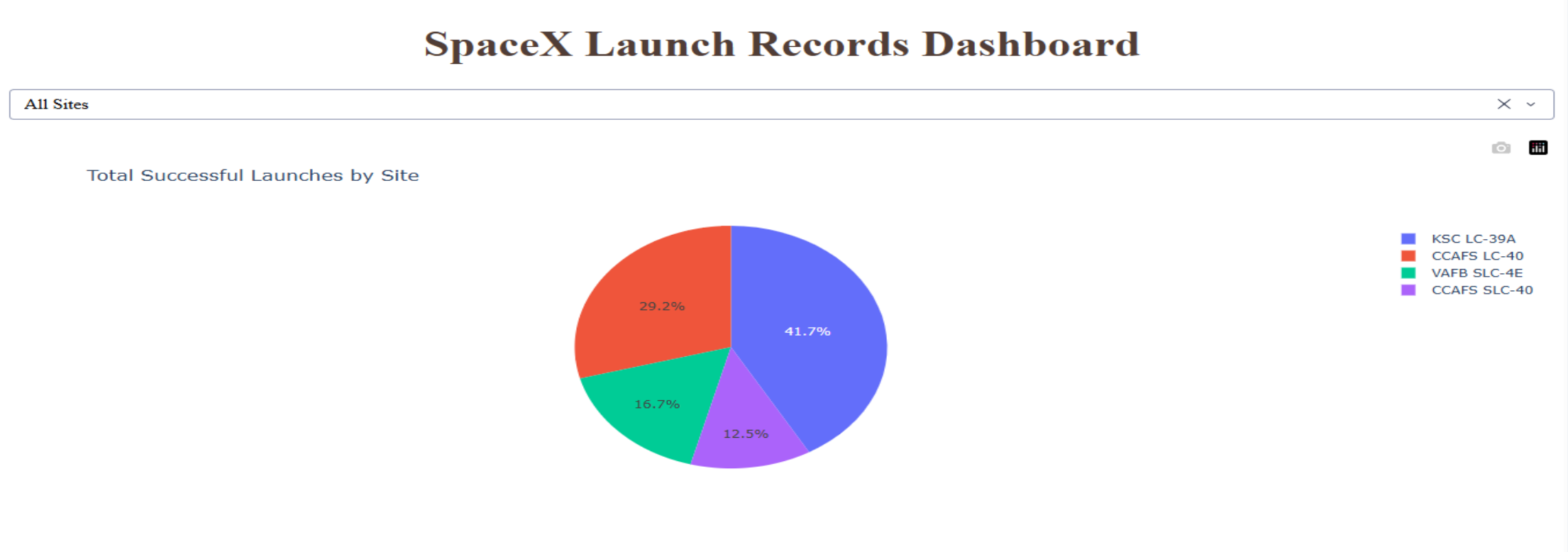


- launch sites are strategically located within **0.5–2.0 km of railways, highways, and coastlines** for logistics and safety, while maintaining a buffer of **over 20 km from cities** to protect populated areas. This positioning ensures easy transport of heavy equipment while allowing flight paths to remain safely over the ocean.

Section 4

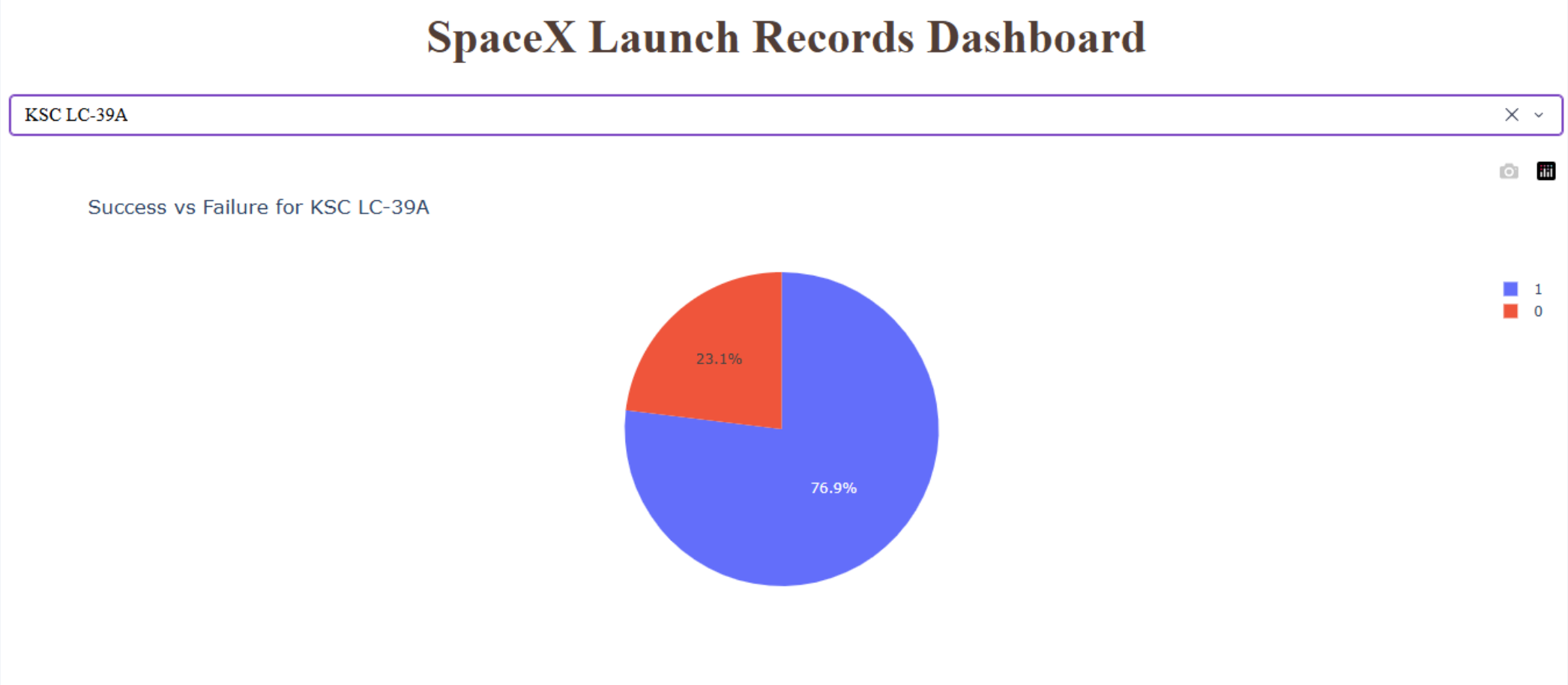
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboards



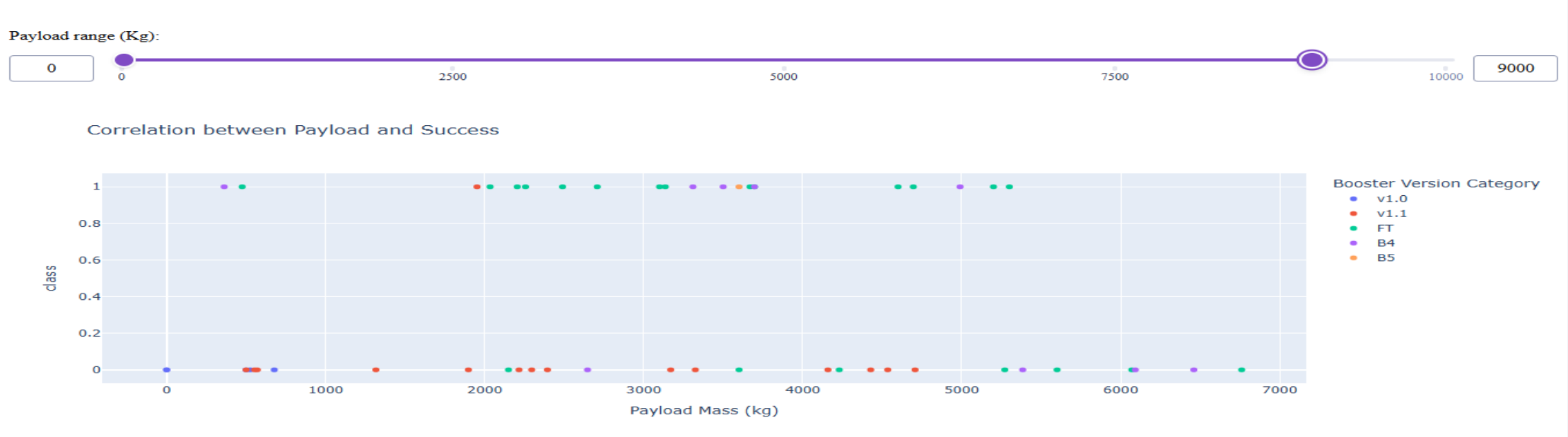
- The dashboard reveals that **KSC LC-39A** is the most successful launch site (41.7%).

KSC LC-39A



- For **KSC LC-39A**, about **76.9% of launches were successful**, while **23.1% failed**, indicating a strong overall success rate at this launch site.

Correlation between payload and success

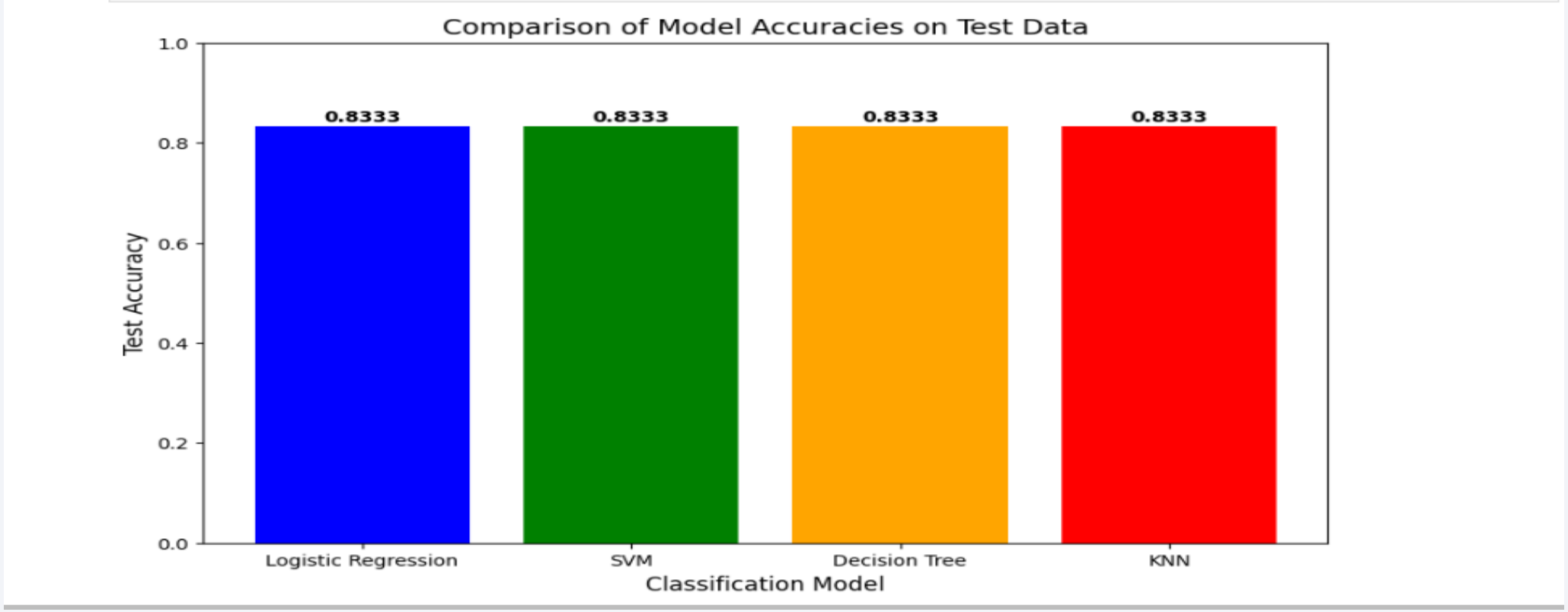


- The dashboard shows no strong direct correlation between payload mass and launch success, as both successes and failures occur across different payload ranges. Booster version appears to influence success more significantly than payload weight alone.

Section 5

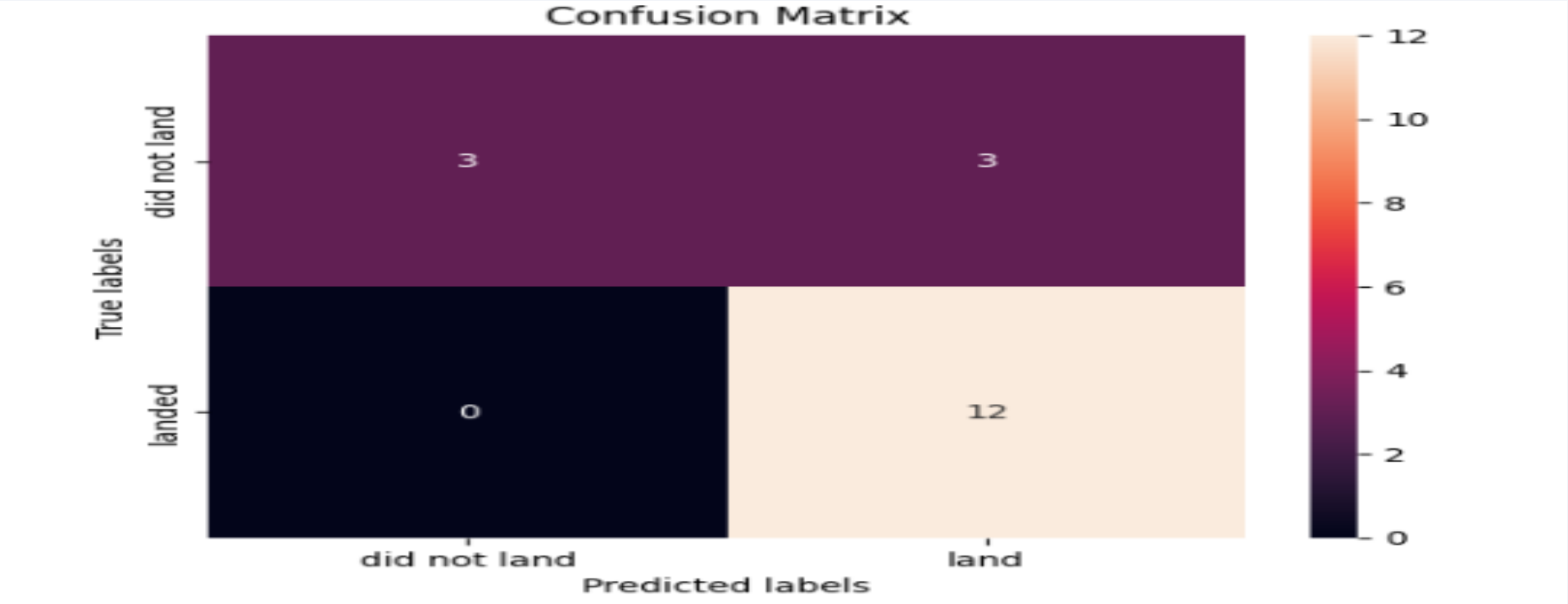
Predictive Analysis (Classification)

Classification Accuracy



- All models have equal model accuracy which is 0.833.

Confusion Matrix



- Examining the confusion matrix of all models, we see that all confusion matrix seems same like as above.
- Overview:
- True Positive - 12 (True label is landed, Predicted label is also landed)
- False Positive - 3 (True label is not landed, Predicted label is landed)

Conclusions

- • Falcon 9 landing success has improved significantly over time due to operational learning and technological advancements.
- • Launch site and booster version strongly influence landing outcomes. **Kennedy Space Center LC-39A** shows the highest success rate, while **Cape Canaveral Space Launch Complex 40** handles high launch volume.
- • Payload mass impacts mission complexity but is not the only determining factor.
- • Classification models achieved **83.3% accuracy**, proving landing success can be effectively predicted using historical data.
- **Final Insight:**
  Data-driven modeling supports better mission planning, risk reduction, and cost optimization.

Appendix

- **Data Sources:-**
- SpaceX REST API
- Web Scraping
- **My GitHub repository:-**
- https://github.com/mukul816/Spacex-project
- **Tools Used:-**
- Python (Pandas, Scikit-learn)
- SQL (SQLite)
- Folium
- Plotly Dash

Models Implemented:-
Logistic Regression
SVM
Decision Tree
KNN
Evaluation Metrics:-
Accuracy (0.833)
Confusion Matrix
Precision, Recall, F1-score

Thank you!